

PROJECT REPORT

Movie Performance Analytics Dashboard

1. INTRODUCTION

The rapid growth of the film industry has led to the generation of large volumes of data related to movie production, revenue, audience engagement, and critical reception. Analyzing this data can provide valuable insights into factors influencing movie success and failure.

This project presents an interactive Movie Analytics Dashboard developed using Power BI, based on data sourced from The Movie Database (TMDB). The dashboard enables the analysis of movie performance across multiple dimensions such as budget, revenue, genre, release period, and audience ratings.

The objective of this project is to transform raw movie data into meaningful visual insights that support data-driven understanding of trends, profitability, and performance outcomes in the movie industry.

2. DATASET DESCRIPTION

The dataset used for this project is derived from The Movie Database (TMDB), a widely used and publicly available movie data platform.

Dataset Attributes Include:

- Budget
- Revenue
- Profit
- Release year and month
- Runtime
- Genre
- Production company
- Vote average and vote count
- Popularity score
- Hit/Flop classification

The dataset was cleaned and pre-processed to remove inconsistencies and ensure analytical accuracy.

Dataset link: <https://developer.themoviedb.org/docs/finding-data>

3. SCOPE OF THE ANALYSIS

The scope of this project includes:

- Analysis of movies released between 1916 and 2016
- Evaluation of financial performance using budget, revenue, and profit
- Classification of movies into Hit and Flop categories
- Analysis of seasonal trends based on release months
- Genre-wise and production company-wise comparisons
- Audience response analysis using vote average and vote count
- Interactive filtering using slicers for dynamic exploration

The project focuses on descriptive and exploratory analysis, rather than predictive modeling.

4. EXISTING SYSTEM

Traditionally, movie performance analysis is carried out using static reports, spreadsheets, or basic charts that offer limited interactivity. These systems often rely on:

- Manual data filtering
- Static tables and charts
- Isolated analysis of individual metrics
- Limited visualization capabilities

Drawbacks / Limitations of Existing System

- Lack of interactivity and dynamic filtering
- Difficulty in identifying trends across multiple dimensions
- Poor scalability with large datasets
- Time-consuming manual analysis
- Limited support for visual storytelling

These limitations make it difficult for stakeholders to gain quick and actionable insights.

5. ANALYSIS ON DATASET

I. INTRODUCTION

This section describes the analytical procedures applied to extract insights from the movie dataset using Power BI.

II. GENERAL DESCRIPTION

The dataset consists of movie-level records where each row represents a single movie. Numerical attributes were used to calculate KPIs, while categorical attributes were used for grouping, filtering, and comparison.

III. SPECIFIC REQUIREMENTS, FUNCTIONS AND FORMULAS

Key DAX measures and calculated columns include:

- **Total Movies**

Total Movies = COUNTROWS(Movies)

- **Average Revenue per Movie**

Avg Revenue per Movie = AVERAGE(Movies[revenue])

- **Average Rating**

Average Rating = AVERAGE(Movies[vote_average])

- **Hit Rate**

Hit Rate (%) =

DIVIDE(

CALCULATE(COUNTROWS(Movies), Movies[hit_flop] = 1),

COUNTROWS(Movies)

)

- **Hit/Flop Label**

Hit Flop Label = IF(Movies[hit_flop] = 1, "Hit", "Flop")

IV. ANALYSIS RESULTS

Key findings from the analysis include:

- A higher concentration of movies released in recent decades
- Seasonal patterns showing higher average revenue for specific release months
- Hit movies significantly outperform flop movies in average revenue
- Certain genres dominate both in production count and revenue contribution
- Production companies vary greatly in output and performance

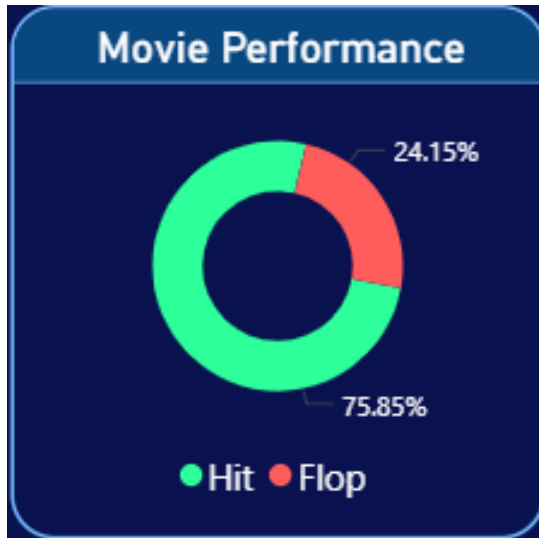
V. VISUALIZATION

The following visualizations were created in Power BI:

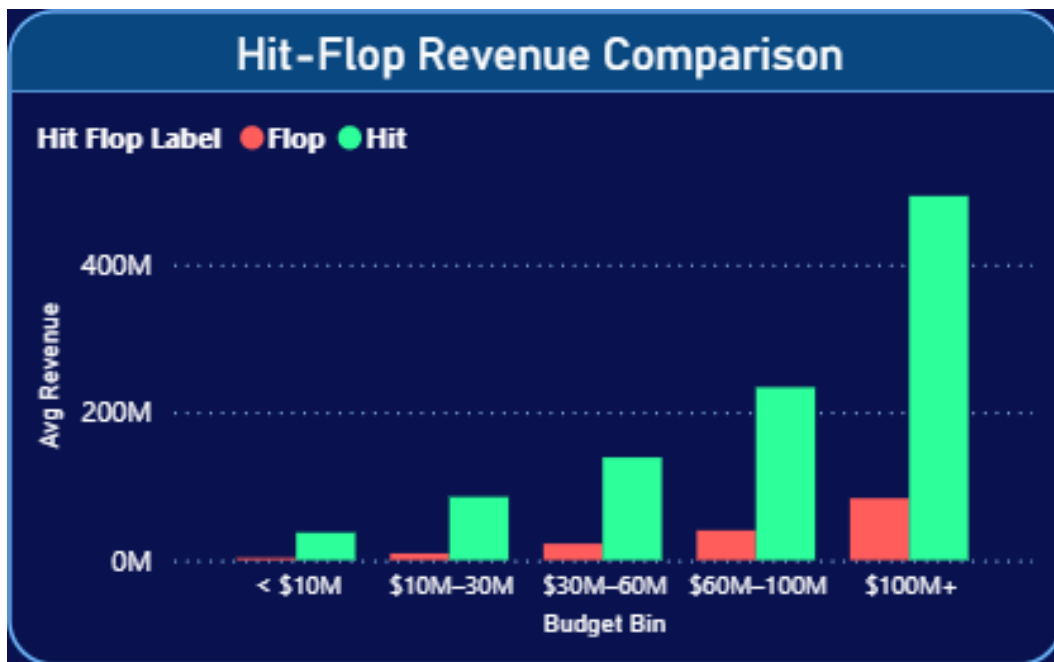
- KPI Cards for summary metrics



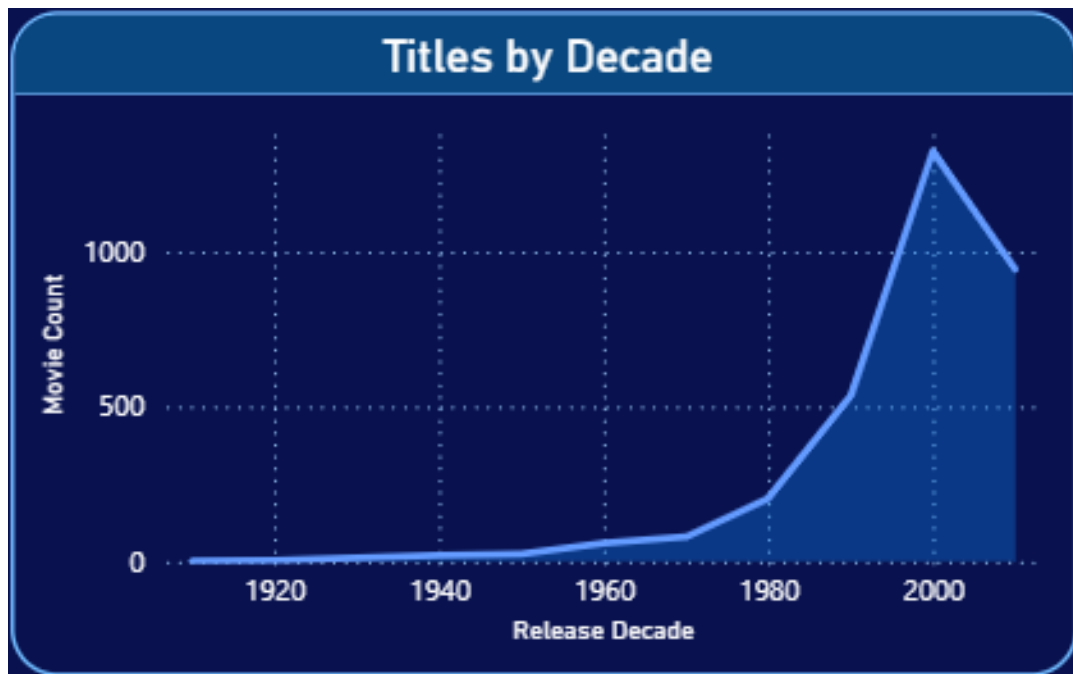
- Donut chart for Hit vs Flop distribution



- Clustered bar charts for revenue comparison

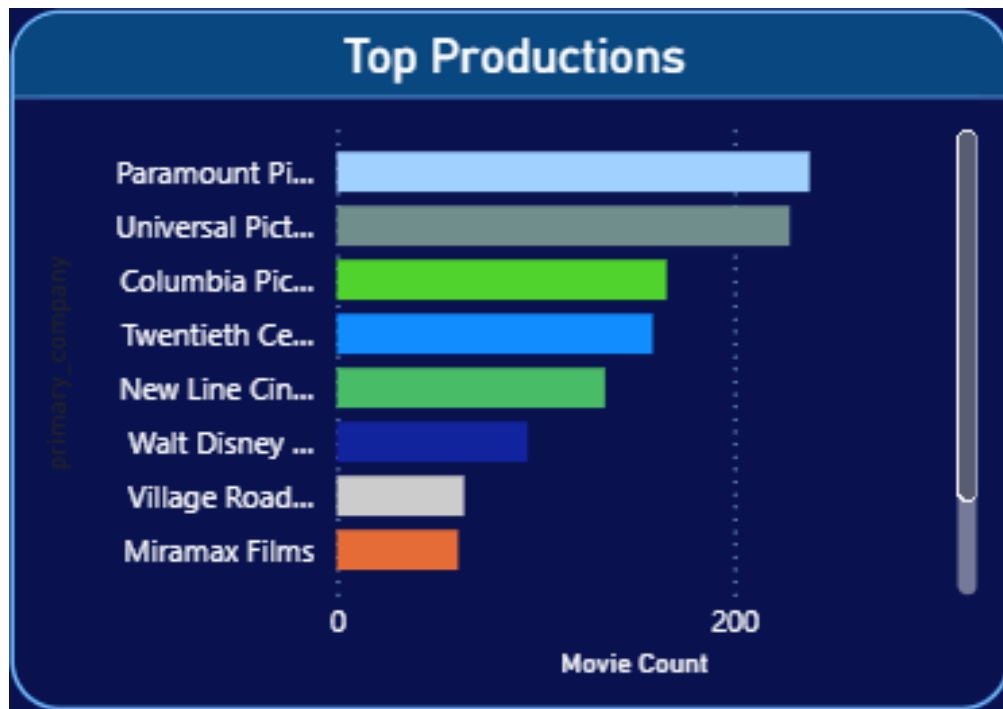


- Area charts for trend analysis



- Genre and production company breakdowns





- Interactive slicers for filtering by year, month, and performance

Release Year

☐ Select...

☐ 2016

☐ 2015

☐ 2014

☐ 2013

☐ 2012

☐ 2011

☐ 2010

Release Month

☐ Select...

☐ Jan

☐ Feb

☐ Mar

☐ Apr

☐ May

☐ Jun

☐ Jul

☐ Hit

☐ Flop

These visuals collectively support effective data exploration and storytelling.

6. List of Analyses with Results

Analysis	Result
Total Movies	Displays dataset size
Genre Distribution	Identifies dominant genres
Hit vs Flop Analysis	Shows performance split
Revenue by Budget Range	Highlights budget efficiency
Monthly Revenue Trend	Reveals seasonal patterns
Decade-wise Releases	Shows industry growth
Production Company Analysis	Identifies major contributors

7. FUTURE SCOPE

Future enhancements to this project may include:

- Implementation of a star schema data model
- Integration of predictive models for hit prediction
- Real-time data integration via APIs
- Sentiment analysis using reviews
- Advanced time intelligence (YoY, MoM analysis)
- Deployment using Power BI Service

8. Conclusion

This project demonstrates how a structured data analytics workflow can transform raw movie industry data into clear, actionable insights through effective business intelligence practices. Using Power BI as the primary analytical tool, the project integrates data preparation, metric engineering, and interactive visualization to analyze movie performance across financial, temporal, and categorical dimensions in an intuitive and user-friendly manner.

From a technical standpoint, the dataset underwent systematic data cleaning, transformation, and feature engineering to ensure consistency and analytical reliability. Key performance indicators were defined using DAX measures, designed to remain responsive to filter context and slicer interactions. Derived metrics such as average revenue, hit rate, and audience ratings were calculated at appropriate levels of data granularity, ensuring meaningful aggregation and comparison across genres, release periods, and performance categories.

To address challenges such as data skewness and overplotting, the analysis adopted aggregation strategies and trend-based visualizations instead of raw point-level representations. Temporal attributes were extracted to support time-series analysis, enabling month-wise and decade-wise trend evaluation. The dashboard design follows established business intelligence and data visualization best practices, including consistent metric definitions, optimized chart selection, and reduced cognitive load through a clean and structured layout.

Overall, this project highlights the practical application of descriptive analytics, exploratory data analysis (EDA), and interactive dashboarding in a real-world context. The resulting dashboard serves as a scalable analytical foundation that can be extended with advanced data modeling, star schema implementation, time intelligence functions, and predictive analytics in future iterations. By balancing technical rigor with clarity and usability, this project reinforces the value of data-driven decision-making and demonstrates the effectiveness of Power BI as a comprehensive business intelligence platform.

9. REFERENCES

1. Microsoft Power BI Documentation
2. The Movie Database (TMDB)
3. Data Visualization Best Practices – Microsoft Learn

10. BIBLIOGRAPHY

- Few, S. (2012). *Show Me the Numbers: Designing Tables and Graphs to Enlighten*.
- Cairo, A. (2016). *The Truthful Art: Data, Charts, and Maps for Communication*.
- TMDB API Documentation

11. LINKS

LinkedIn: https://www.linkedin.com/posts/alwinw_powerbi-dataanalytics-businessintelligence-activity-7406052614519500800-Ig_3

GitHub: <https://github.com/AlwinW64/Movie-Performance-Analytics-Dashboard>